BREATHE INDIA

The air India breathes

26 cities · 2015 – mid-2020 · CPCB monitoring network

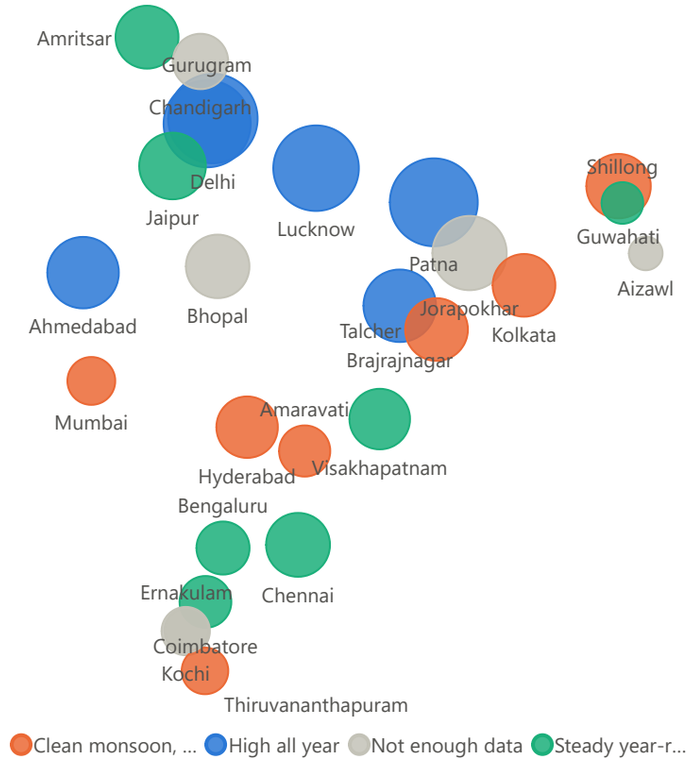
Overview | City explorer | Lockdown | Early warning | Data quality

Cities monitored	City-days with PM2.5 data	Typical PM2.5 (µg/m³)	Days above PM2.5 limit
26	25K	48	39%

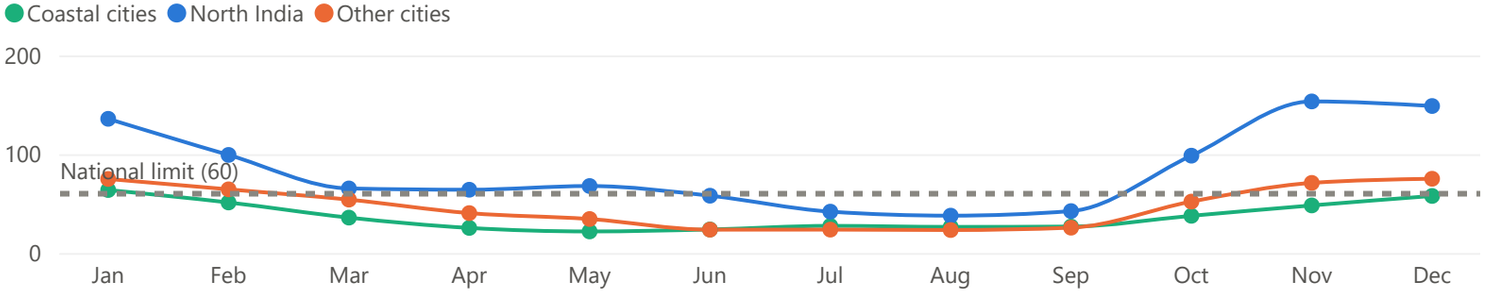
Year: All

Region: All

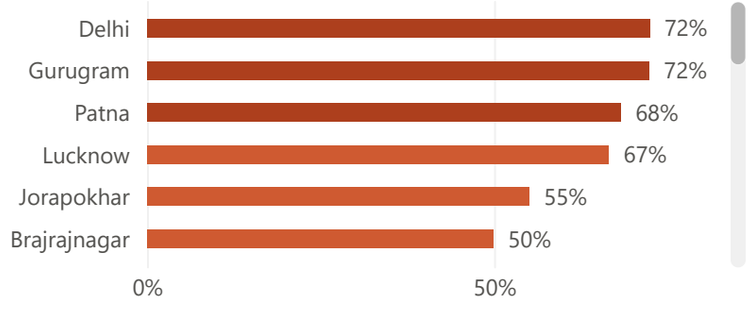
Where the air is worst (bubble = typical PM2.5)



North India's winter air is about 3x dirtier than its monsoon air



Delhi and Gurugram break the limit on 7 days in 10



What this page says

Across 26 cities, PM2.5 broke India's daily limit on **39% of measured days.**

The worst air is in the north (Delhi, Gurugram, Patna, Lucknow) and it peaks in **November–January**, when crop burning, cold still air and festival fireworks combine.

Hover a city for its profile. Use City explorer for the detail.

Data: Central Pollution Control Board (via Kaggle) · Open-Meteo / Copernicus ERA5 (CC BY 4.0)

Mahak Wadhawan (L074) · Yash Shah (L056)

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City explorer

Pick a city: its seasons, its stations and its worst days

Overview | City... | Lockdown | Early warning | Data quality

Each city's seasonal fingerprint (median PM2.5) · click a city to filter

City	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Ahmedabad	76	78	72	78	46	40	42	41	36	90	90	72
Aizawl			35	22	10	2	4					
Amaravati	76	42	27	18	25	18	14	14	26	25	58	77
Amritsar	56	49	41	34	44	40	39	31	38	63	88	70
Bengaluru	43	42	42	35	29	21	20	18	22	33	35	43
Bhopal	70	49	31	41	20	16	14		16	51	91	79
Brajrajnagar	95	81	93	64	51	39	25	30	22	32	79	83
Chandigarh	46	45	19	13	20	22	33		27	56	68	69
Chennai	57	51	50	44	46	46	47	38	33	41	45	52
Coimbatore	36	40	33	23	24	17	33	22	22	24	38	38
Delhi	190	141	85	89	87	65	44	43	55	132	210	209
Ernakulam	48	43	28	17	16	10	6					
Gurugram	151	110	73	83	94	79	55	41	49	109	161	169
Guwahati	152	115	95	46	24	21	25	22	24	46	64	119
Hyderabad	67	55	49	41	37	22	18	18	28	54	63	64
Jaipur	63	54	42	45	51	47	36	32	34	67	77	63
Jorapokhar	74	84	67	50	47	27		14	37	50	83	85
Kochi	63	54	36	18	17	17	6					
Kolkata	122	99	59	31	22	21	24	20	20	63	106	121
Lucknow	180	128	89	80	73	64	36	38	41	117	198	193
Mumbai	73	61	29	21	15	14	14	14	19	44	63	71
Patna	203	122	92	64	58	47	40	42	50	107	211	243
Shillong	27	49	32	34	14	10	17	18	9	18	12	7

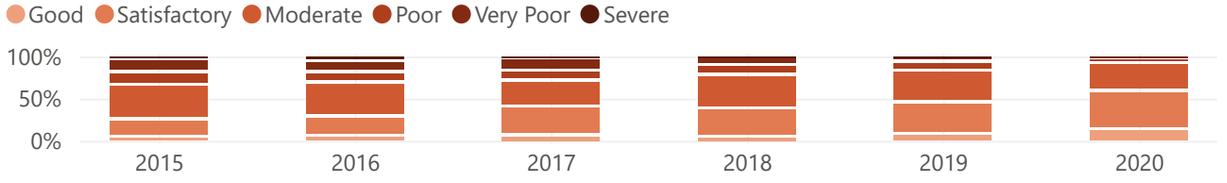
City

All

All 26 cities: 39% of days over the PM2.5 limit · worst in Nov

Typical AQI	Days Poor or worse	Usually sets the AQI	Worst month
113	22%	PM2.5	Nov

How many days fell in each AQI band, by year



PM2.5, 7-day average



Worst monitoring stations (mean PM2.5)

Station	PM2.5
Anand Vihar	153
Talkatora District Industries Center	135
DTU	131

Data: Central Pollution Control Board (via Kaggle) · Open-Meteo / Copernicus ERA5 (CC BY 4.0)
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What the lockdown really changed

Actual pollution vs. what the weather alone would have produced

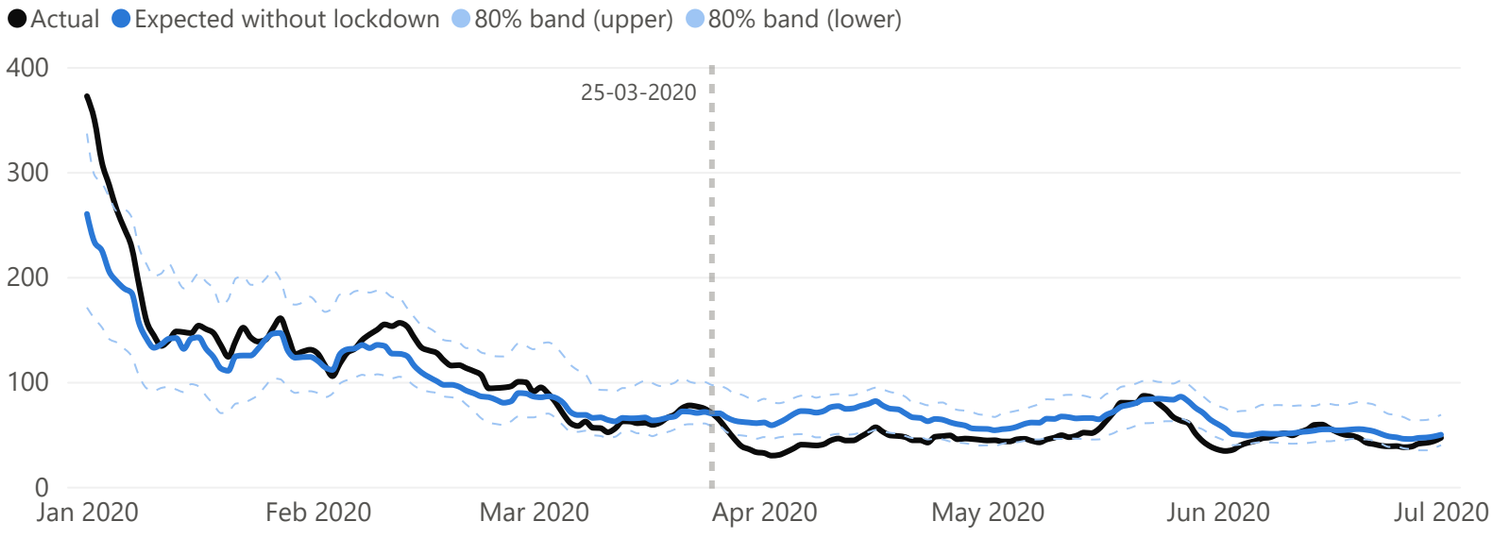
Overview | City explorer | **Lockdown** | Early warning | Data quality

PM2.5 vs 2019	PM2.5, weather removed	Drop due to weather	NO_2, weather removed
-44%	-28%	35%	-49%

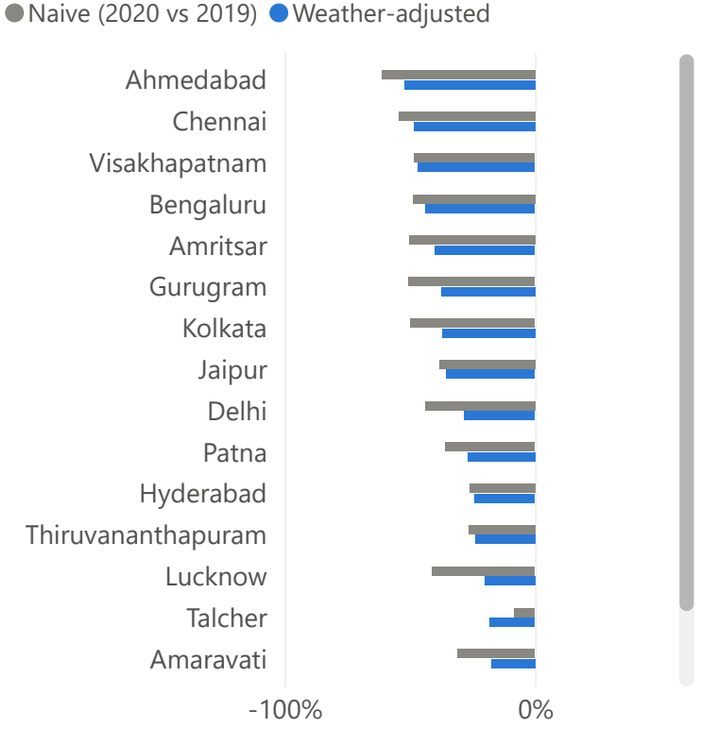
City: Delhi

Pollutant: PM25

Actual vs expected without lockdown, 2020 (pick a city and pollutant)



PM2.5 change by city: weather made most drops look bigger



Comparing spring 2020 with spring 2019 says PM2.5 fell **41%**. But spring 2020 also had kinder weather. Our model, trained on 2015–2019, predicts what pollution should have been given the actual weather. Against that, the real lockdown effect was **−28%**: about **a third of the headline drop was weather**. NO_2, mostly from traffic, fell **46%** either way.

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Can we warn a day ahead?

Next-day AQI category model vs. a 'same as today' baseline

Model: exact category	Baseline ('same as today')	Poor-or-worse days caught	Right when the category changes
75%	72%	92%	35%

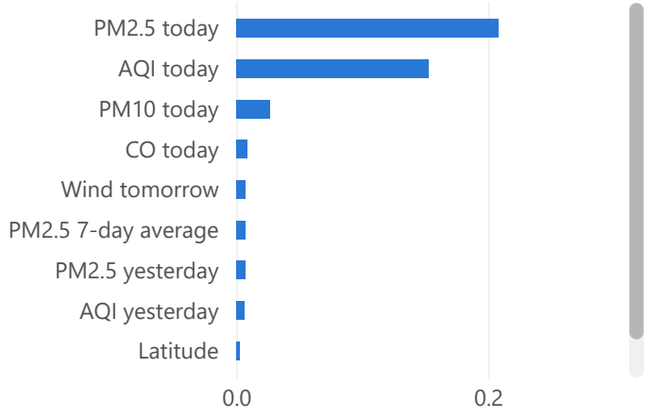
Evaluation period

Test

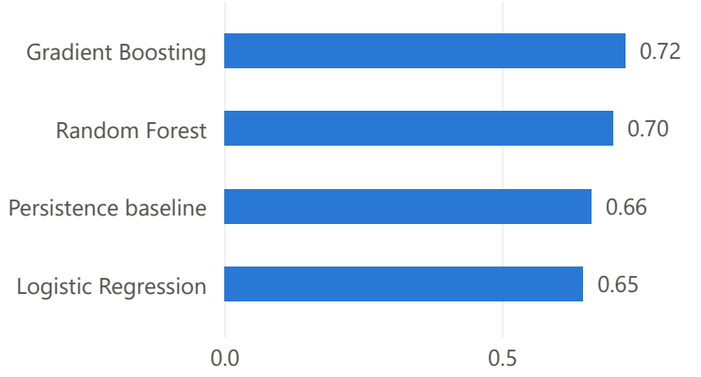
Where predictions land (row % of actual days, test period)

Actual	Good	Satisfactory	Moderate	Poor	Very Poor	Severe
Good	74%	26%	0%	0%	0%	0%
Satisfactory	9%	79%	12%	0%	0%	0%
Moderate	0%	15%	72%	12%	0%	0%
Poor	0%	0%	13%	76%	11%	0%
Very Poor	0%	0%	0%	20%	73%	6%
Severe	0%	0%	2%	1%	20%	77%

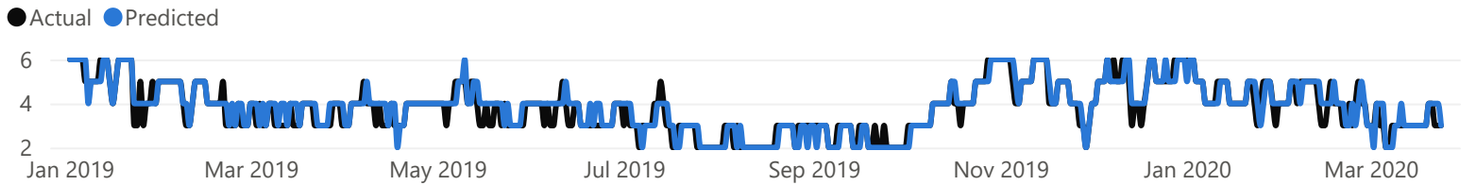
What the model relies on (top 10 inputs)



Macro F1 on unseen 2019–20 data: Gradient Boosting beats the baseline



Delhi: predicted vs actual AQI category (1 = Good ... 6 = Severe)



The model gets **75%** of next-day categories exactly right (baseline 72%) and catches **92% of Poor-or-worse days** (baseline 80%).

Honest caveat: switch the period to Lockdown and it does worse than the baseline, because the rules it learned stopped applying.

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How the data was cleaned

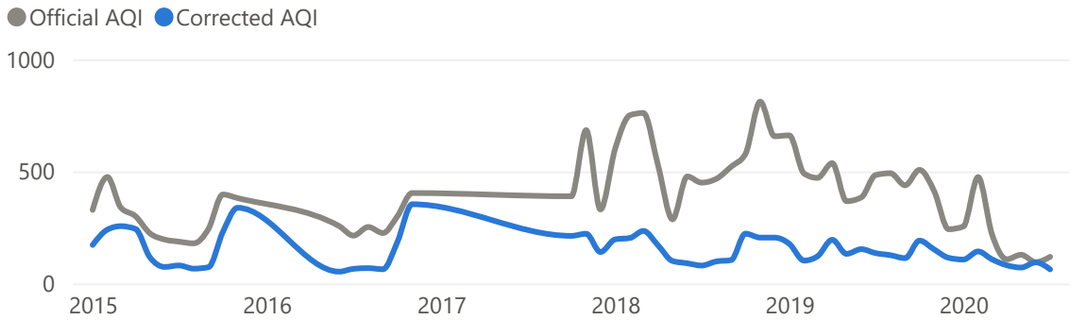
What we found, fixed and flagged before any analysis

Cleaning decisions logged	Sensor faults removed	AQI values rebuilt
23	408	3K

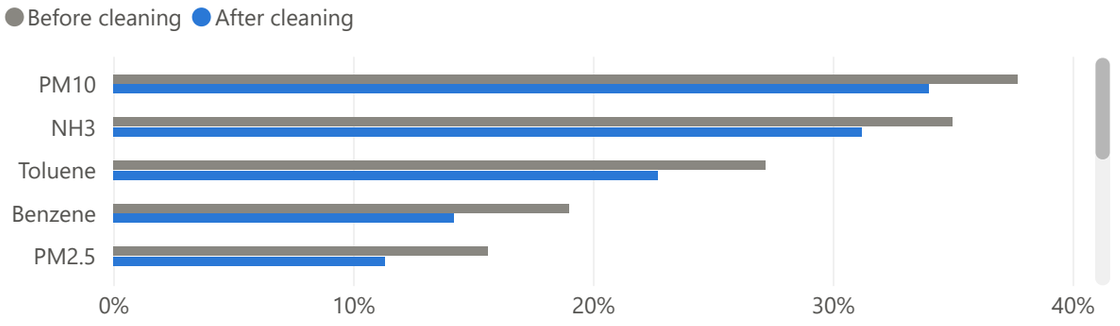
Ahmedabad: #1 most polluted → #7 after cleaning

Its CO sensor column was a copy of NO (identical on 94% of days).

Ahmedabad: official AQI vs corrected (monthly median)



Missing values before vs after cleaning



Cleaning log: every decision and why

#	Table	Issue found	Action	Reason
1	city_day	Date stored as text	Parsed to datetime with an explicit format (0 failures)	Needed for joins on date, lags and seasonal features
2	all	Possible duplicate rows or keys	Checked all tables; none found. Uniqueness is enforced again by the database keys	Keys must be unique for one-to-one joins
3	city_day	Ahmedabad CO equals NO on most days (r = 0.999) although units differ 1000x	Set all Ahmedabad CO values to missing and flagged the rows	Corrupted CO also inflates the official AQI; see the AQ
4	city_day	Exact 0.00 daily means for CO and NOx	Set to missing	A city-wide daily average of exactly zero is impossible
5	city_day	Exact zeros in Benzene / Toluene / Xylene	Kept as-is	Plausible 'below detection limit' readings, and these cc

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